

Run: rk4-wout-300e

Log-log loss curves

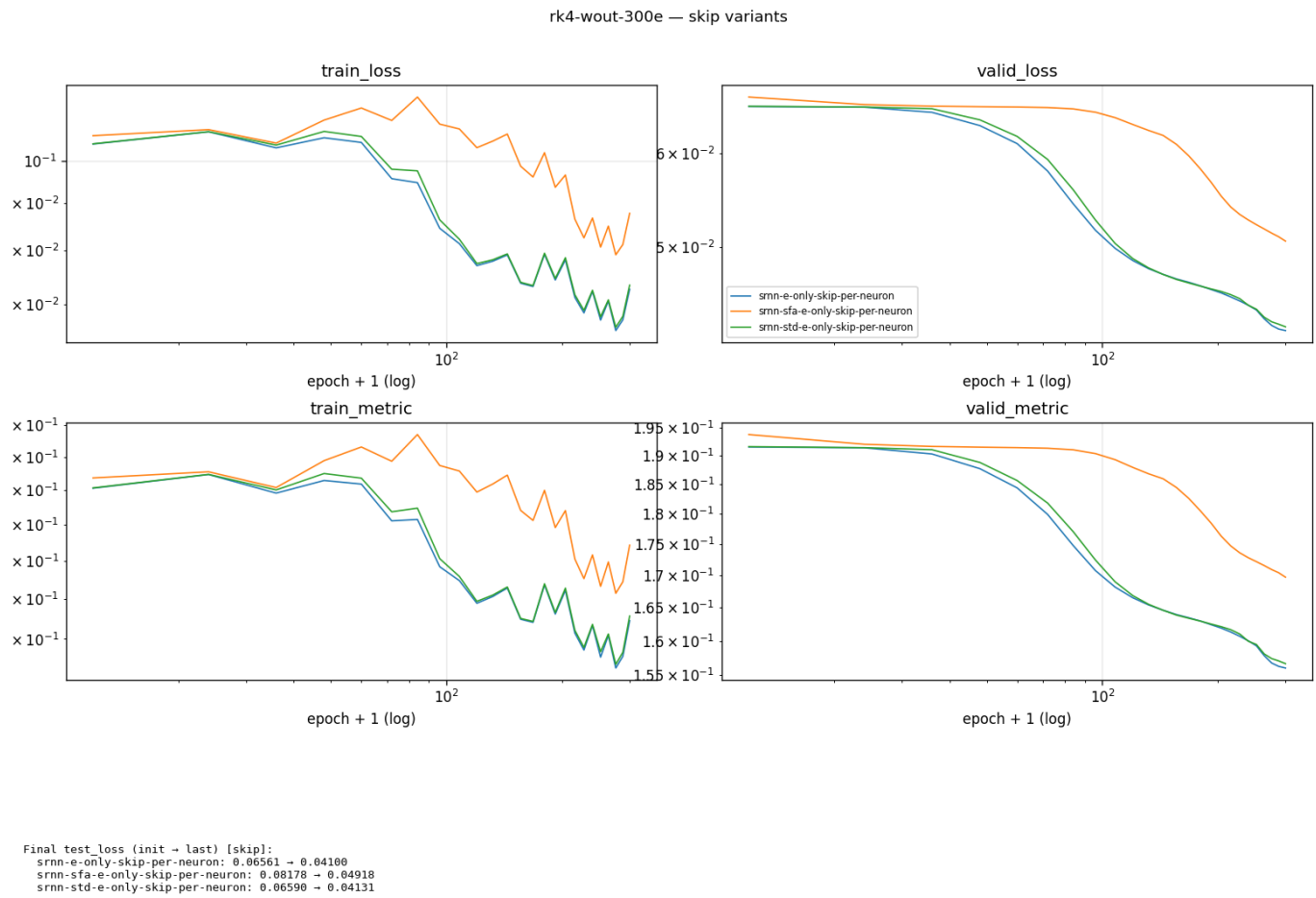
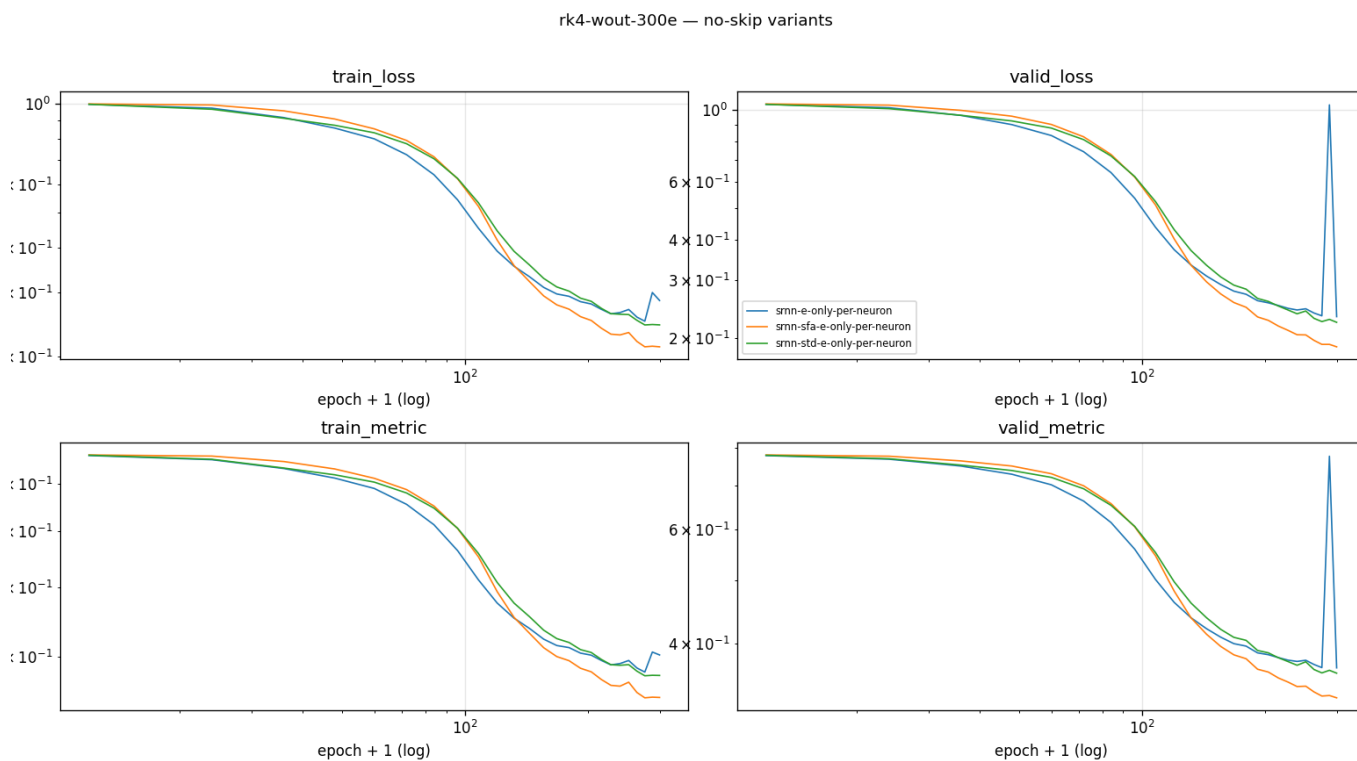


Figure 1: skip variants — log-log loss/metric



Final test loss (init - last) [no-skip]:
 srnn-e-only-per-neuron: 0.90042 - 0.21153
 srnn-sfa-e-only-per-neuron: 0.91618 - 0.17300
 srnn-std-e-only-per-neuron: 0.90077 - 0.20387

Figure 2: no-skip variants — log-log loss/metric

srnn-e-only-per-neuron

rk4-wout-300e — effective time constants (srnn-e-only-per-neuron)

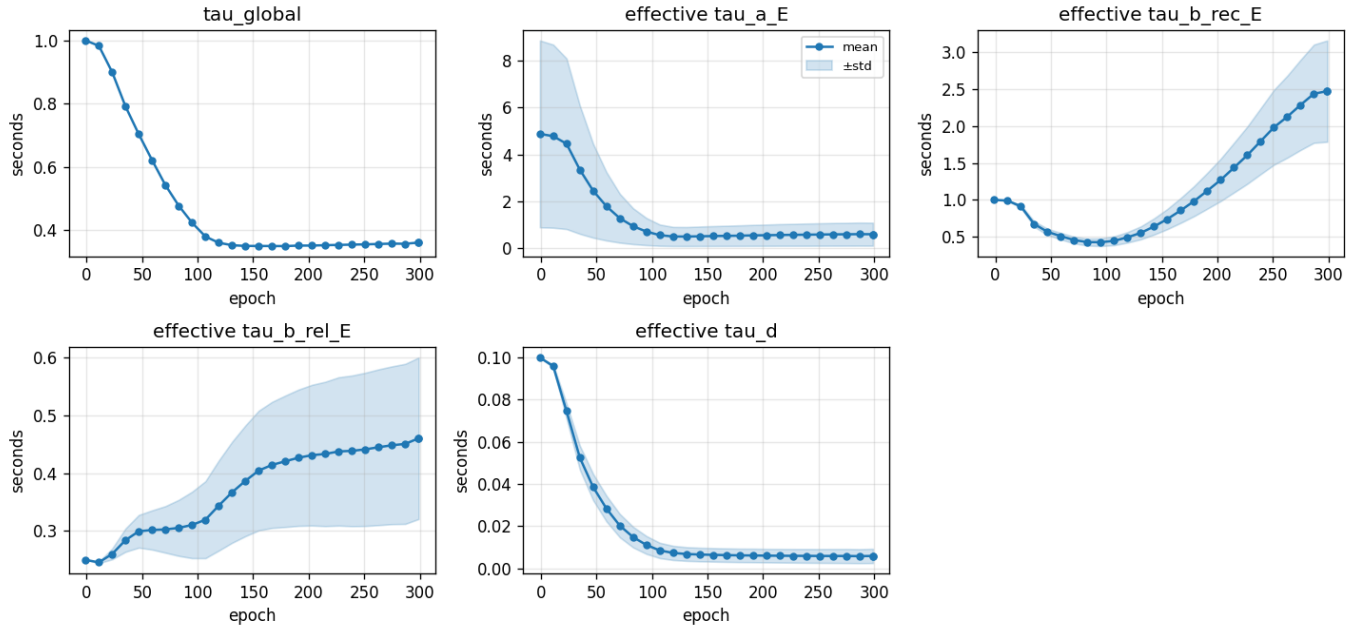
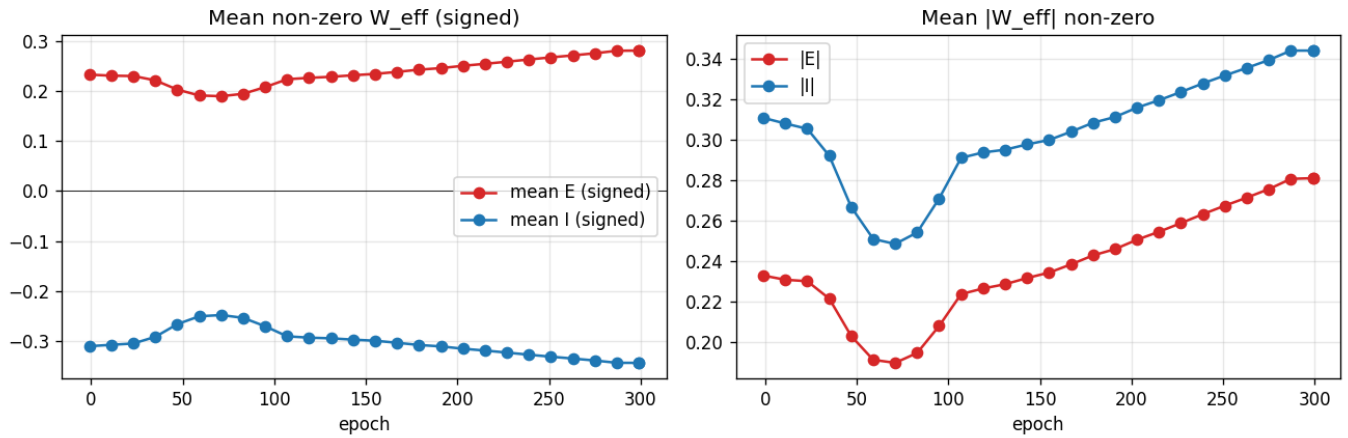


Figure 3: Tau evolution

rk4-wout-300e — recurrent E vs I weights (srnn-e-only-per-neuron)

Figure 4: W_{EI} evolution

srnn-e-only-per-neuron — parameter table

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# rk4-wout-300e - variant srnn-e-only-per-neuron (k=0)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.36117	-	-	scalar
tau_d (s)	+0.1	+0.0059435	0	0.003387	300
tau_a_E[j=0] (s)	+0.25	+0.014364	0	0.006425	150
tau_a_E[j=1] (s)	+4.3832	+0.53873	0	0.08523	150
tau_a_E[j=2] (s)	+10	+1.2049	0	0.07775	150
tau_b_rec_E (s)	+1	+2.4697	0	0.6851	150
tau_b_rel_E (s)	+0.25	+0.46061	0	0.1399	150
W_eff (E src, signed)	+0.23267	+0.28082	0.07641	0.1368	15048
W_eff (I src, signed)	-0.31062	-0.34389	0.07724	0.1101	14977
W_eff (E src)	+0.23267	+0.28082	0.07641	0.1368	15048
W_eff (I src)	+0.31062	+0.34389	0.07724	0.1101	14977
W_in_eff (input neurons)	-0.0003386	+6.718e-05	0.1012	0.04455	6675
c_E[j=0] (SFA coupling)	+0.05	+0.32175	0	0.2115	150
c_E[j=1] (SFA coupling)	+0.05	+0.13592	0	0.03189	150
c_E[j=2] (SFA coupling)	+0.05	+0.14416	0	0.03062	150
c_0_E[j=0] (SFA offset)	+0	-0.7737	0	0.1655	150
c_0_E[j=1] (SFA offset)	+0	-1.0014	0	0.3186	150
c_0_E[j=2] (SFA offset)	+0	-1.0468	0	0.3779	150
a_0 (threshold)	+0.35	-0.82632	0	0.1613	300
readout_weight	-2.8854e-05	+0.0014493	0.06604	0.3687	6675
readout_bias	+0	+0.023748	0	0.1002	89
ic.ic (per-variant init)	+0.10233	+0.10233	0.3832	0.3832	900

srnn-e-only-skip-per-neuron

rk4-wout-300e — effective time constants (srnn-e-only-skip-per-neuron)

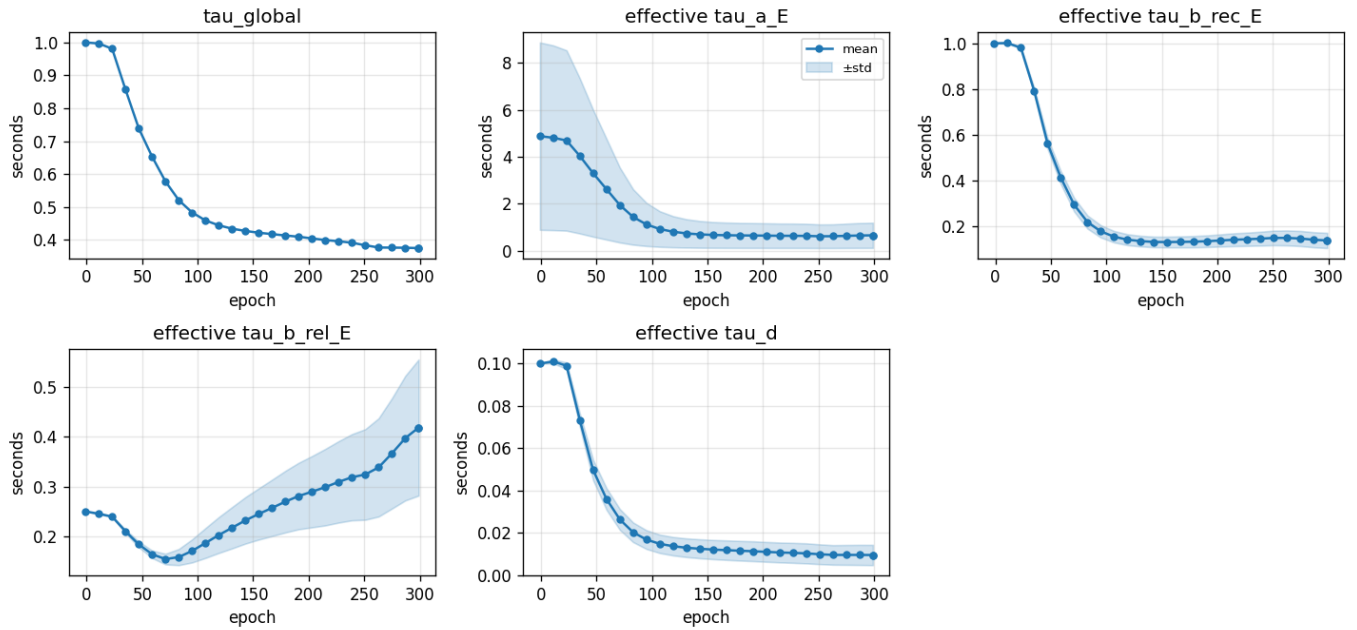
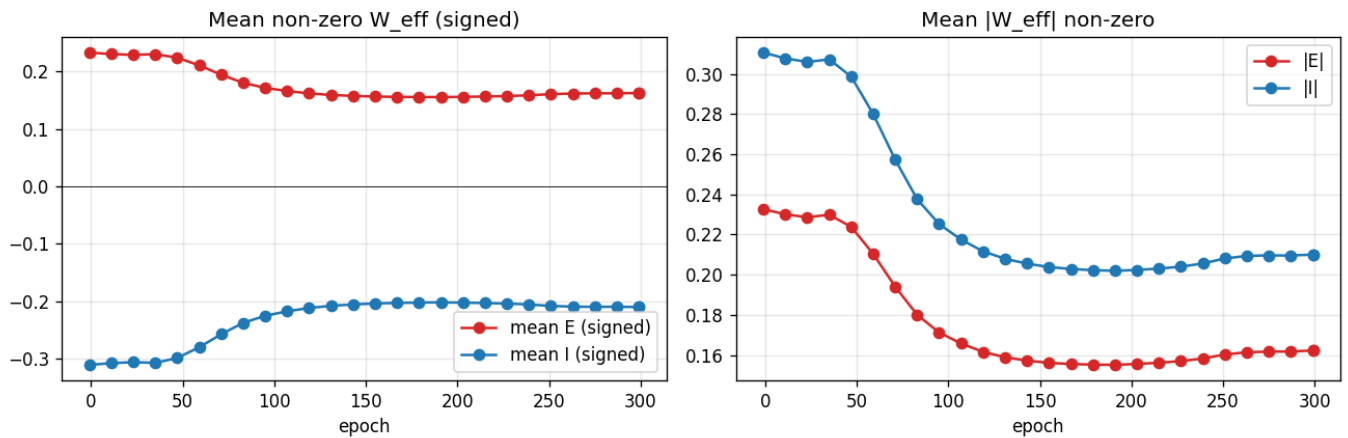


Figure 5: Tau evolution

rk4-wout-300e — recurrent E vs I weights (srnn-e-only-skip-per-neuron)

Figure 6: W_{EI} evolution

srnn-e-only-skip-per-neuron — parameter table

rk4-wout-300e - variant srnn-e-only-skip-per-neuron (k=1)
Effective values (post-transform). Means/stds computed over the indicated dimension.

effective parameter	init mean	final mean	init std	final std	n
tau_global (s)	+1	+0.37529	-	-	scalar
tau_d (s)	+0.1	+0.0095989	0	0.004796	300
tau_a_E[j=0] (s)	+0.25	+0.022312	0	0.01249	150
tau_a_E[j=1] (s)	+4.3832	+0.59099	0	0.05747	150
tau_a_E[j=2] (s)	+10	+1.3304	0	0.0452	150
tau_b_rec_E (s)	+1	+0.13967	0	0.03327	150
tau_b_rel_E (s)	+0.25	+0.4179	0	0.1362	150
W_eff (E src, signed)	+0.23267	+0.1623	0.07641	0.06233	15048
W_eff (I src, signed)	-0.31062	-0.21008	0.07724	0.06407	14977
W_eff (E src)	+0.23267	+0.1623	0.07641	0.06233	15048
W_eff (I src)	+0.31062	+0.21008	0.07724	0.06407	14977
W_in_eff (input neurons)	-0.00084237	-0.0003466	0.1008	0.01914	6675
c_E[j=0] (SFA coupling)	+0.05	+0.14153	0	0.07546	150
c_E[j=1] (SFA coupling)	+0.05	+0.082642	0	0.01972	150
c_E[j=2] (SFA coupling)	+0.05	+0.085951	0	0.01753	150
c_0_E[j=0] (SFA offset)	+0	-0.37156	0	0.1258	150
c_0_E[j=1] (SFA offset)	+0	-0.416	0	0.2162	150
c_0_E[j=2] (SFA offset)	+0	-0.41887	0	0.2292	150
a_0 (threshold)	+0.35	-0.057435	0	0.1178	300
readout_weight	+0.00078219	-0.0061936	0.06654	0.1652	6675
readout_bias	+0	+0.12832	0	0.1694	89
ic.ic (per-variant init)	+0.10092	+0.10092	0.3832	0.3832	900

srnn-sfa-e-only-per-neuron

rk4-wout-300e — effective time constants (srnn-sfa-e-only-per-neuron)

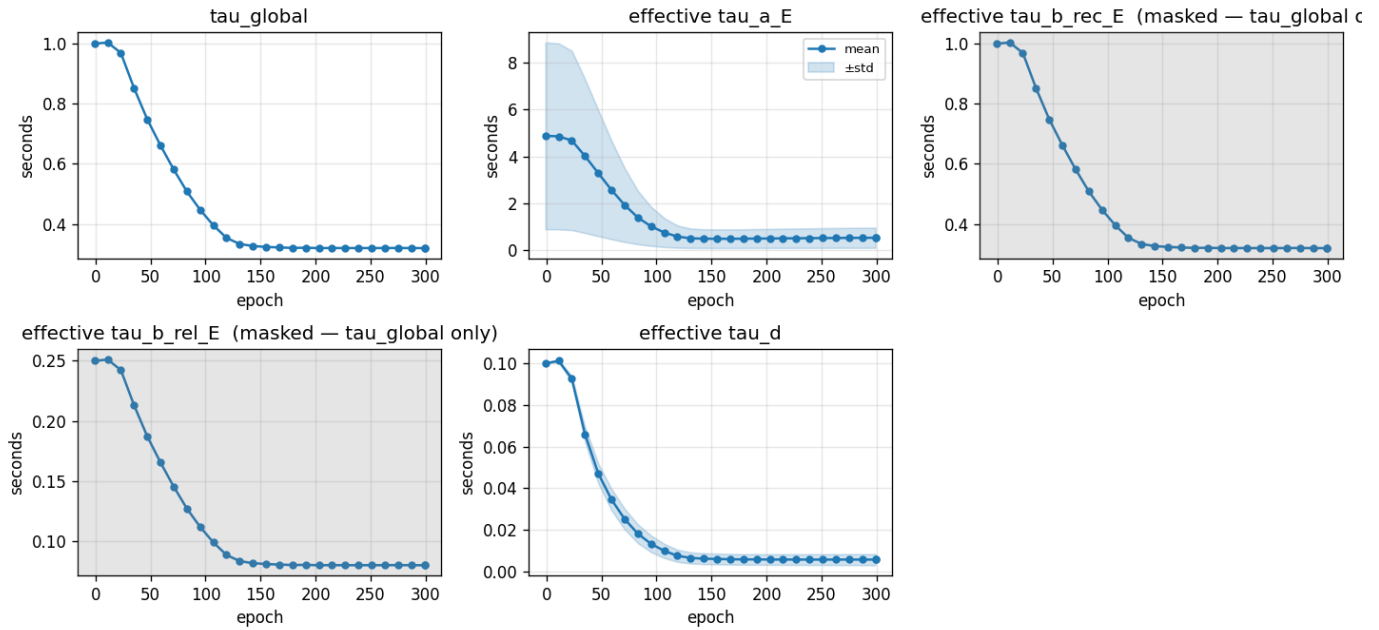
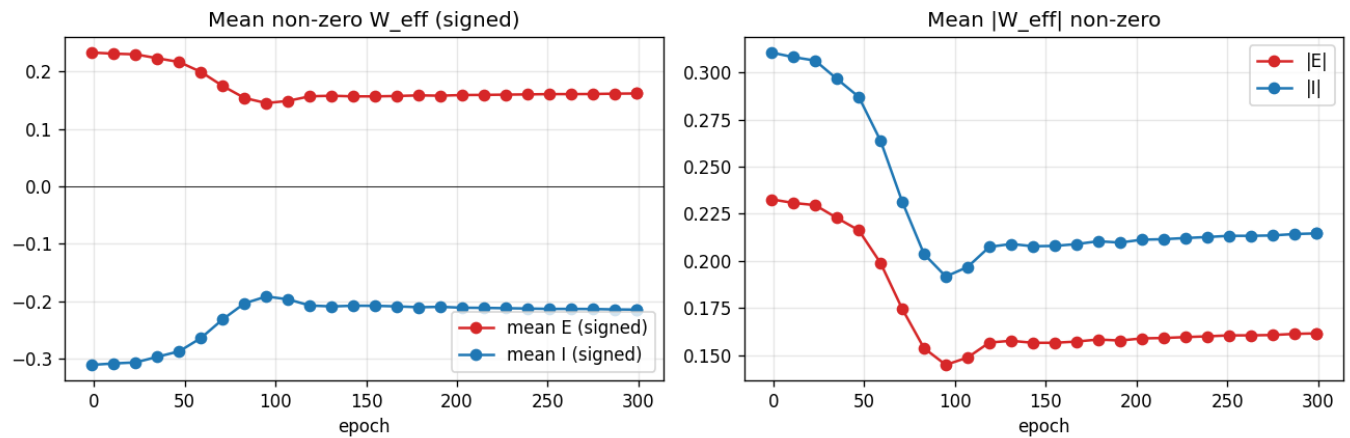


Figure 7: Tau evolution

rk4-wout-300e — recurrent E vs I weights (srnn-sfa-e-only-per-neuron)

Figure 8: W_{EI} evolution

srnn-sfa-e-only-per-neuron — parameter table

rk4-wout-300e - variant srnn-sfa-e-only-per-neuron (k=2)
Effective values (post-transform). Means/stds computed over the indicated dimension.

effective parameter	init mean	final mean	init std	final std	n
tau_global (s)	+1	+0.3198	-	-	scalar
tau_d (s)	+0.1	+0.005601	0	0.002688	300
tau_a_E[j=0] (s)	+0.25	+0.013493	0	0.006264	150
tau_a_E[j=1] (s)	+4.3832	+0.47568	0	0.06381	150
tau_a_E[j=2] (s)	+10	+1.0598	0	0.06605	150
tau_b_rec_E (s)	+1	+0.3198	0	0	150
tau_b_rel_E (s)	+0.25	+0.07995	0	0	150
W_eff (E src, signed)	+0.23267	+0.16148	0.07641	0.0616	15048
W_eff (I src, signed)	-0.31062	-0.21457	0.07724	0.06388	14977
W_eff (E src)	+0.23267	+0.16148	0.07641	0.0616	15048
W_eff (I src)	+0.31062	+0.21457	0.07724	0.06388	14977
W_in_eff (input neurons)	-0.00015382	-0.00067693	0.09934	0.044	6675
c_E[j=0] (SFA coupling)	+0.05	+0.45524	0	0.3174	150
c_E[j=1] (SFA coupling)	+0.05	+0.17414	0	0.05186	150
c_E[j=2] (SFA coupling)	+0.05	+0.18888	0	0.04269	150
c_0_E[j=0] (SFA offset)	+0	-0.39567	0	0.179	150
c_0_E[j=1] (SFA offset)	+0	-0.40352	0	0.2373	150
c_0_E[j=2] (SFA offset)	+0	-0.40179	0	0.2497	150
a_0 (threshold)	+0.35	+0.25068	0	0.1195	300
readout_weight	-0.00098972	+0.003701	0.06678	0.2512	6675
readout_bias	+0	-0.06826	0	0.1322	89
ic.ic (per-variant init)	+0.098719	+0.098719	0.4415	0.4415	900

srnn-sfa-e-only-skip-per-neuron

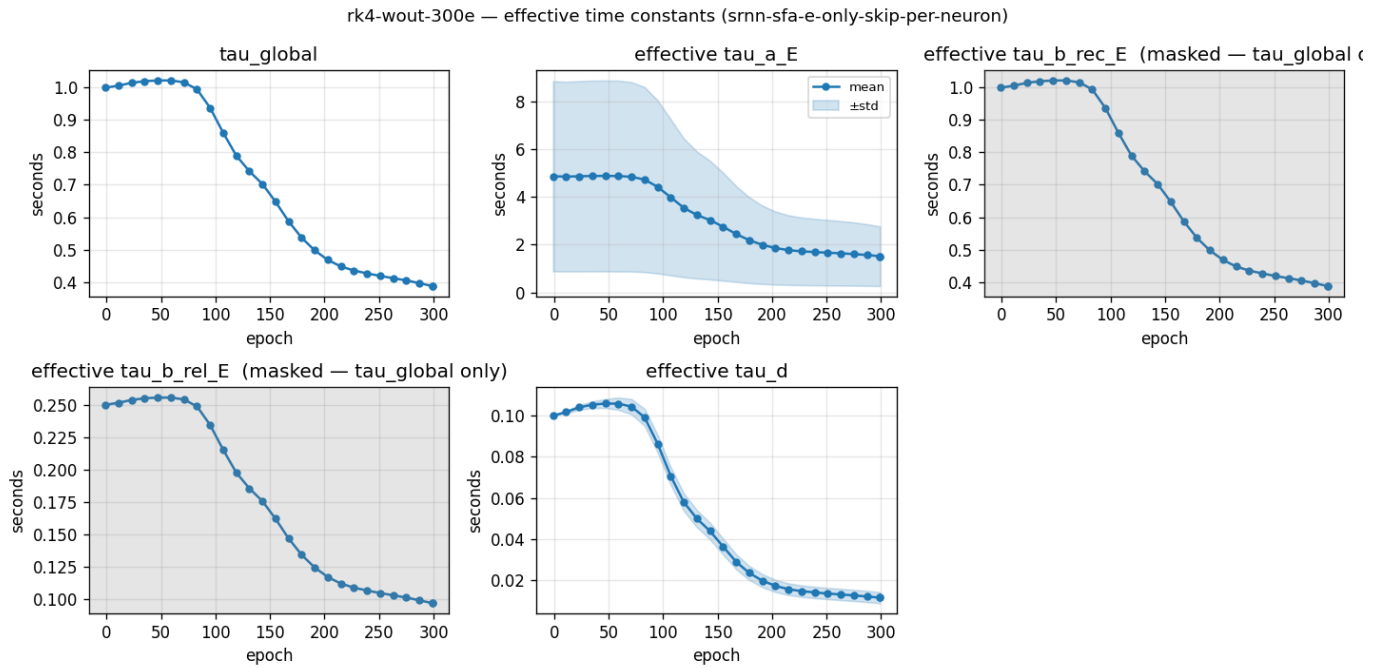
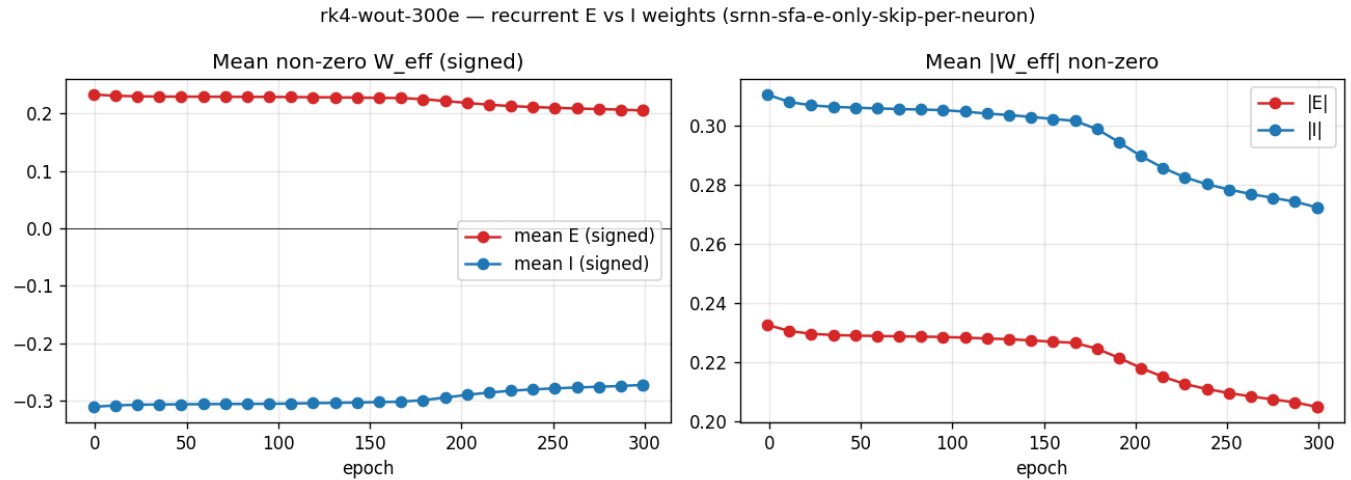


Figure 9: Tau evolution

Figure 10: W_{EI} evolution

srnn-sfa-e-only-skip-per-neuron — parameter table

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# rk4-wout-300e - variant srnn-sfa-e-only-skip-per-neuron (k=3)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.38706	-	-	scalar
tau_d (s)	+0.1	+0.011607	0	0.00269	300
tau_a_E[j=0] (s)	+0.25	+0.070031	0	0.01582	150
tau_a_E[j=1] (s)	+4.3832	+1.3521	0	0.05551	150
tau_a_E[j=2] (s)	+10	+3.1278	0	0.02274	150
tau_b_rec_E (s)	+1	+0.38706	0	0	150
tau_b_rel_E (s)	+0.25	+0.096764	0	0	150
W_eff (E src, signed)	+0.23267	+0.20496	0.07641	0.07074	15048
W_eff (I src, signed)	-0.31062	-0.27248	0.07724	0.07366	14977
W_eff (E src)	+0.23267	+0.20496	0.07641	0.07074	15048
W_eff (I src)	+0.31062	+0.27248	0.07724	0.07366	14977
W_in_eff (input neurons)	-0.0011621	-0.00046372	0.09962	0.03058	6675
c_E[j=0] (SFA coupling)	+0.05	+0.060161	0	0.02322	150
c_E[j=1] (SFA coupling)	+0.05	+0.053451	0	0.01059	150
c_E[j=2] (SFA coupling)	+0.05	+0.053542	0	0.01022	150
c_0_E[j=0] (SFA offset)	+0	-0.05172	0	0.1646	150
c_0_E[j=1] (SFA offset)	+0	-0.049636	0	0.1857	150
c_0_E[j=2] (SFA offset)	+0	-0.048053	0	0.1822	150
a_0 (threshold)	+0.35	+0.28874	0	0.1135	300
readout_weight	+0.00016801	-0.0040284	0.0673	0.1052	6675
readout_bias	+0	+0.076065	0	0.1312	89
ic.ic (per-variant init)	+0.12557	+0.12557	0.42	0.42	900

srnn-std-e-only-per-neuron

rk4-wout-300e — effective time constants (srnn-std-e-only-per-neuron)

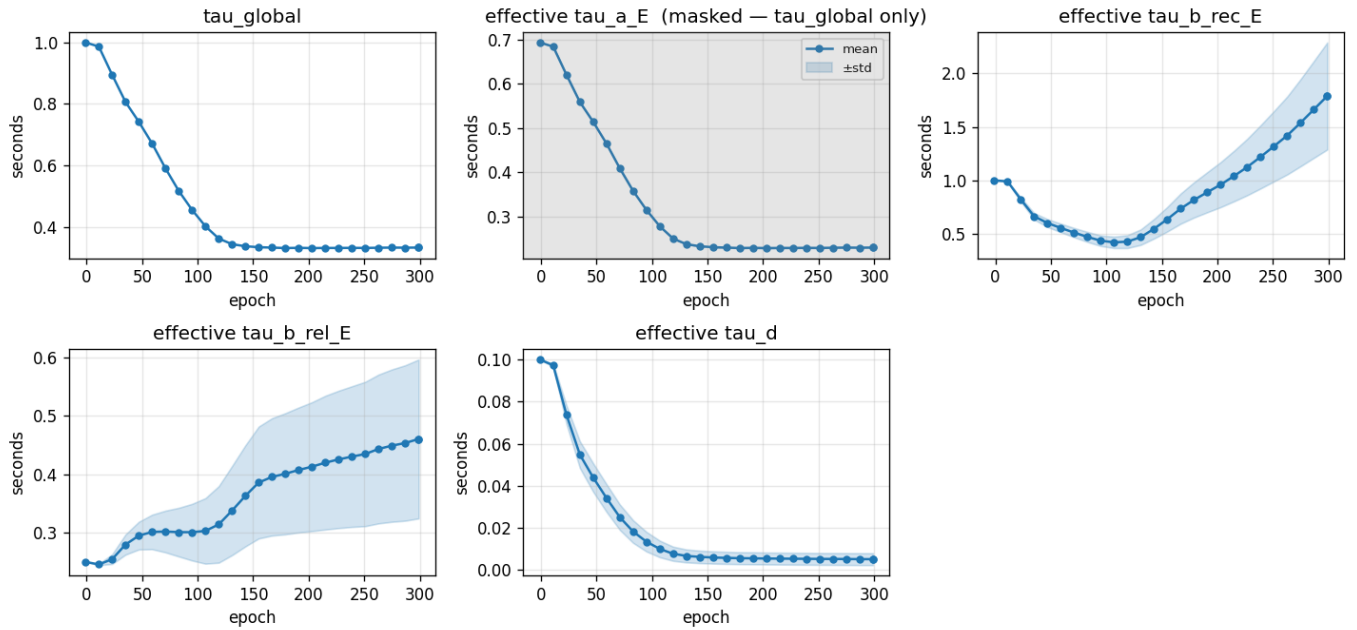
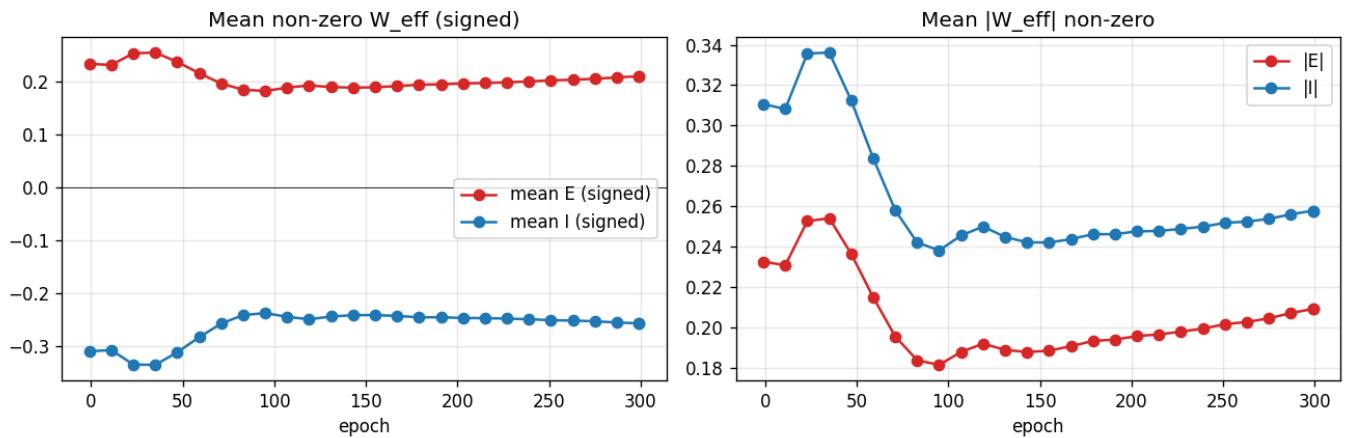


Figure 11: Tau evolution

rk4-wout-300e — recurrent E vs I weights (srnn-std-e-only-per-neuron)

Figure 12: W_{EI} evolution

srnn-std-e-only-per-neuron — parameter table

```
# rk4-wout-300e - variant srnn-std-e-only-per-neuron (k=4)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.33251	-	-	scalar
tau_d (s)	+0.1	+0.0051415	0	0.002853	300
tau_b_rec_E (s)	+1	+1.7897	0	0.5016	150
tau_b_rel_E (s)	+0.25	+0.46015	0	0.1359	150
W_eff (E src, signed)	+0.23267	+0.20924	0.07641	0.0974	15048
W_eff (I src, signed)	-0.31062	-0.25784	0.07724	0.08169	14977
W_eff (E src)	+0.23267	+0.20924	0.07641	0.0974	15048
W_eff (I src)	+0.31062	+0.25784	0.07724	0.08169	14977
W_in_eff (input neurons)	+0.00019944	-6.4633e-07	0.09994	0.03827	6675
a_0 (threshold)	+0.35	-0.85975	0	0.1836	300
readout_weight	+0.00074661	-0.00094686	0.06657	0.3627	6675
readout_bias	+0	-0.080185	0	0.2094	89
ic.ic (per-variant init)	+0.090198	+0.090198	0.3825	0.3825	900

srnn-std-e-only-skip-per-neuron

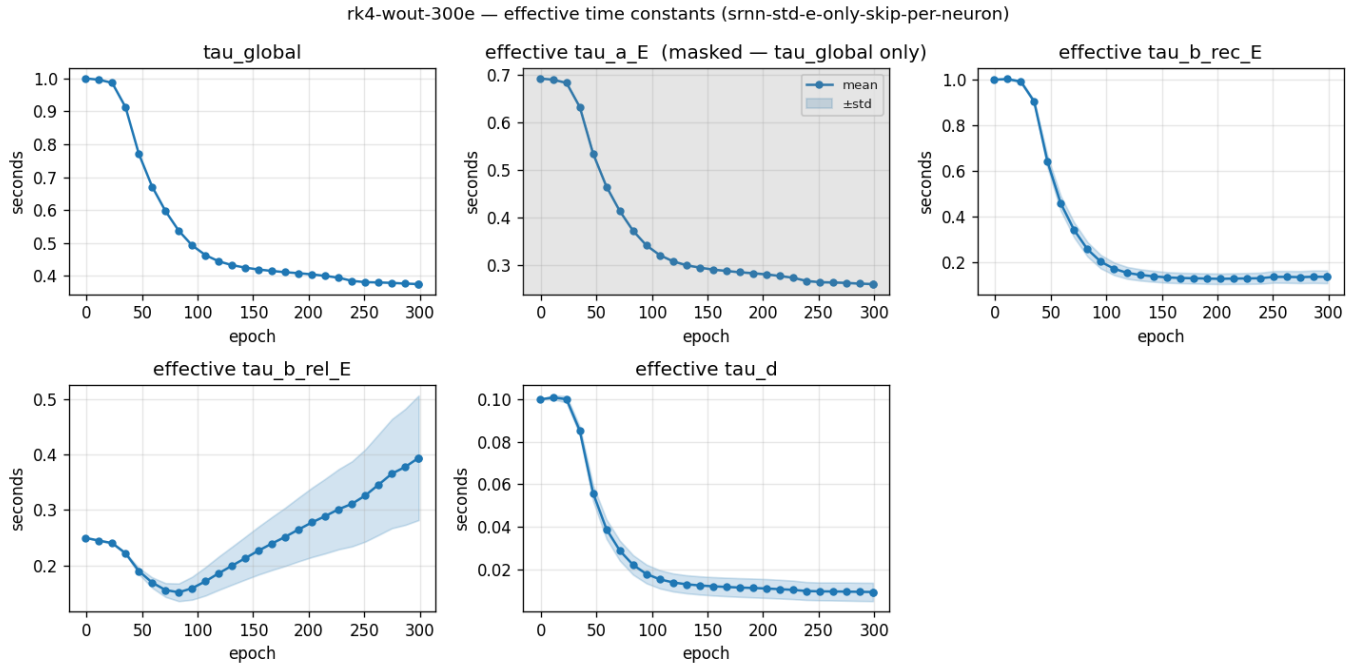
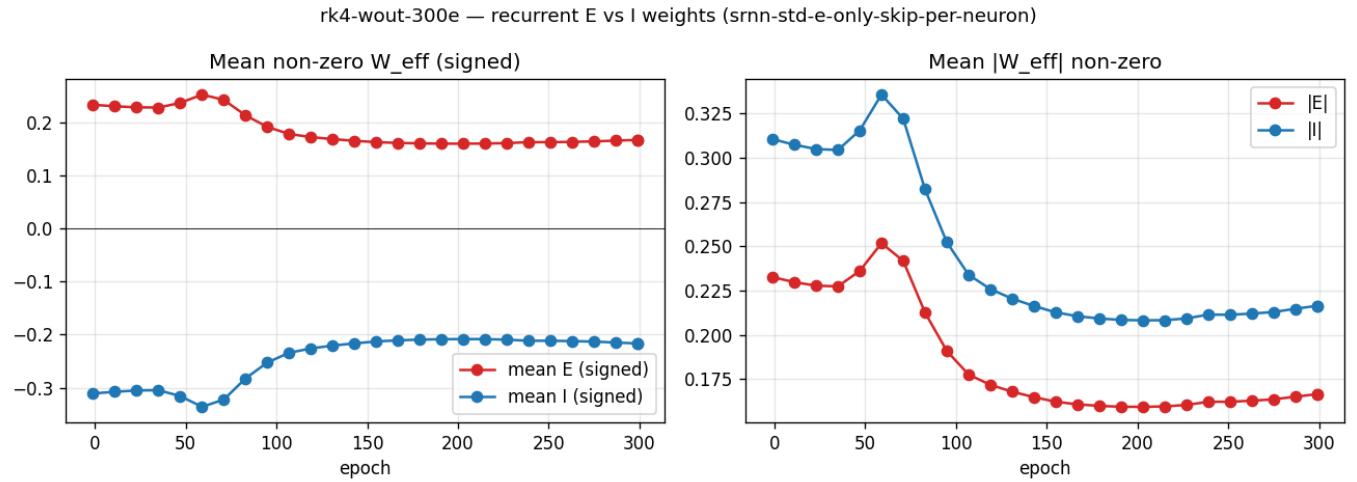


Figure 13: Tau evolution

Figure 14: W_{EI} evolution

srnn-std-e-only-skip-per-neuron — parameter table

```
# rk4-wout-300e - variant srnn-std-e-only-skip-per-neuron (k=5)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n
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tau_global (s)	+1	+0.37503	-	-	scalar
tau_d (s)	+0.1	+0.0092527	0	0.004334	300
tau_b_rec_E (s)	+1	+0.13503	0	0.02779	150
tau_b_rel_E (s)	+0.25	+0.39409	0	0.1122	150
W_eff (E src, signed)	+0.23267	+0.16661	0.07641	0.06346	15048
W_eff (I src, signed)	-0.31062	-0.21652	0.07724	0.06415	14977
W_eff (E src)	+0.23267	+0.16661	0.07641	0.06346	15048
W_eff (I src)	+0.31062	+0.21652	0.07724	0.06415	14977
W_in_eff (input neurons)	+0.00036119	+0.00025925	0.09956	0.01927	6675
a_0 (threshold)	+0.35	-0.031354	0	0.1263	300
readout_weight	+0.00055372	-0.0050808	0.06572	0.1581	6675
readout_bias	+0	+0.030615	0	0.1208	89
ic.ic (per-variant init)	+0.082727	+0.082727	0.3836	0.3836	900